

ASHKAN SHAHBAZI

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Education

Vanderbilt University

Nashville, TN, USA

Ph.D. Candidate, Computer Science

Aug 2022 – Present (Expected 2026)

- Advisor: [Prof. Soheil Kolouri](#)
- Committee: S. Kolouri, M. I. Miga, J. Y. Wu, T. Derr

Amirkabir University of Technology

Tehran, Iran

B.Sc., Biomedical Engineering

Sept 2016 – Jun 2020

- Thesis: ML-based cuffless blood pressure estimation from biosignals

Machine Learning Summer School, Columbia University

New York, NY, USA

Admitted Participant (with grant)

Summer 2026

Research Interests

Developing principled, geometrically-grounded architectures for scalable deep learning—with a focus on attention mechanisms, LLMs and neural simulation.

Professional Experiences

Machine Learning Engineer, Ph.D. Intern

Sep 2025 – Nov 2025

Instacart 🍌, Content AI Team

Remote (San Francisco, CA)

Mentor: [Shih-Ting Lin](#)

- Designed and deployed an LLM cascade for catalog attribute extraction, reducing per-item inference cost by ~75–80% (to ~20–25% of an always-o3 pipeline) while matching or slightly improving o3 F1 across 50+ attributes on ~150k labeled products.
- Benchmarked three confidence estimation families (self-consistency, cross-LLM agreement, verifier ensembles) on a 147k-sample internal golden set to guide routing decisions.
- Trained a lightweight student verifier to predict 5-level confidence via distillation from cross-model agreement and verifier ensembles, improving F1 and cost-aware cascade metrics at a fraction of ensemble compute.

Selected Publications (* EQUAL CONTRIBUTION)

Conference Papers

1. **A. Shahbazi**, E. Akbari, D. Salehi, X. Liu, N. Naderializadeh, and S. Kolouri (2025). **ESPFormer: Doubly Stochastic Attention with Expected Sliced Transport Plans**. *International Conference on Machine Learning (ICML)*.
2. **A. Shahbazi***, K. Pereira*, J. S. Heiselman, E. Akbari, A. C. Benson, S. Seifi, X. Liu, G. L. Johnston, J. Y. Wu, N. Simaan, M. I. Miga, and S. Kolouri (2026). **Neural Augmented Kelvinlet for Real-Time Soft Tissue Deformation Modeling**. *AAAI Conference on Artificial Intelligence (AAAI)*.
3. X. Liu, Y. Bai, R. D. Martín, K. Shi, **A. Shahbazi**, B. A. Landman, C. Chang, and S. Kolouri (2025). **Linear Spherical Sliced Optimal Transport: A Fast Metric for Comparing Spherical Data**. *International Conference on Learning Representations (ICLR), SPOTLIGHT*.
4. X. Liu, R. D. Martín, Y. Bai, **A. Shahbazi**, M. Thorpe, A. Aldroubi, and S. Kolouri (2025). **Expected Sliced Transport Plans**. *International Conference on Learning Representations (ICLR)*.

5. H. Tran*, Y. Bai*, A. Kothapalli*, **A. Shahbazi**, X. Liu, R. D. Martín, and S. Kolouri (2024). **Stereographic Spherical Sliced Wasserstein Distances.** *International Conference on Machine Learning (ICML), SPOTLIGHT.*

Workshop Papers

1. A. Kothapalli, **A. Shahbazi**, X. Liu, R. Sheng, and S. Kolouri (2024). **Equivariant vs Invariant Layers: A Comparison of Backbone and Pooling for Point Cloud Classification.** *ICML Workshop on Geometry-Grounded Representation Learning and Generative Modeling (GRaM).*
2. A. Abbasi, **A. Shahbazi**, H. Pirsiavash, and S. Kolouri (2026). **One Category One Prompt: Dataset Distillation Using Diffusion Models.** *Synthetic Data for Computer Vision Workshop (SynData4CV), CVPR 2026.*

Preprints

1. **A. Shahbazi***, E. Akbari*, K. Pereira, J. S. Heiselman, A. C. Benson, G. L. Johnston, J. Y. Wu, N. Simaan, M. I. Miga, and S. Kolouri (2026). **SurgFormer: Scalable Learning of Organ Deformation with Resection Support and Real-Time Inference.** arXiv:2603.06543.
2. **A. Shahbazi***, P. He*, A. Abbasi, Y. Bai, X. Liu, E. Akbari, N. Naderializadeh, and S. Kolouri (2025). **LUNA: Linear Universal Neural Attention with Generalization Guarantees.** arXiv:2512.08061.
3. **A. Shahbazi**, C. Thrash, Y. Bai, K. Hamm, N. Naderializadeh, and S. Kolouri (2025). **LOTFormer: Doubly Stochastic Linear Attention via Low-Rank Optimal Transport.** arXiv:2509.23436.
4. **A. Shahbazi**, X. Liu, P. He, and S. Kolouri (2025). **MinGSGW: A Scalable Path to Gromov–Wasserstein.** arXiv:2605.13753
5. H. Tran*, Y. Bai*, **A. Shahbazi***, J. R. Hershey, and S. Kolouri (2024). **Understanding Learning with Sliced Wasserstein Requires Rethinking Informative Slices.** arXiv:2411.10651.
6. A. Abbasi, **A. Shahbazi**, H. Pirsiavash, and S. Kolouri (2025). **Vector Quantized Soft Label Compression for Dataset Distillation.** arXiv:2603.03808.

US Patents

- **A. Shahbazi**, K. Pereira, J. S. Heiselman, E. Akbari, A. C. Benson, S. Kolouri, M. I. Miga, and N. Simaan (2025). **System and Method for Real-Time Soft Tissue Simulation via Hybrid AI-Driven Physics-Informed Modeling.** U.S. Patent Application No. 63/998,405.

Selected Research Experience

Attention Mechanisms and Large Language Models (*Ongoing*)

- Developed ESPFormer, a doubly stochastic attention mechanism that replaces iterative Sinkhorn normalization with a single parallelizable step: 1D OT plans between projected query/key distributions are computed and lifted to form an Expected Sliced Transport Plan (ESTP), yielding a doubly stochastic coupling with tunable sparsity via an inverse temperature parameter. (*ICML 2025*)
- Developed LOTFormer, which achieves $\mathcal{O}(n)$ doubly stochastic attention by constraining the transport coupling to be low-rank. Two entropic OT problems are solved between queries/keys and a small learnable pivot measure of support size $r \ll n$. The resulting attention matrix is provably doubly stochastic, rank- r , and applies to values in $\mathcal{O}(nr)$ time. (*arXiv 2025*)

- Developed LUNA, a kernelized linear attention that eliminates the accuracy–efficiency tradeoff of prior linear methods by learning the kernel feature map itself, while enabling $\mathcal{O}(n)$ time and memory. Achieves state-of-the-art accuracy among efficient Transformers on the Long Range Arena and supports post-hoc linearization of pretrained softmax models. (*arXiv 2025*)
- Deployed an LLM cascade for catalog attribute extraction at Instacart: confidence-guided routing via self-consistency, cross-LLM agreement, and verifier ensembles over a 147k-sample golden set; distilled student verifier cutting cost $\sim 75\text{--}80\%$ vs. always-o3 while matching F1 across 50+ attributes on $\sim 150\text{k}$ products. (*Instacart, 2025*)

Optimal Transport Theory and Applications (*Ongoing*)

- Co-developed the Expected Sliced Transport Plan (ESTP) framework, which constructs couplings by averaging 1D OT plans over random projections on the unit sphere, producing an explicit ambient-space transport plan provably forming a valid metric on discrete measures; serves as the theoretical foundation for ESPFormer. (*ICLR 2025*)
- Co-developed linear and stereographic spherical sliced Wasserstein distances on $\mathbb{S}^{d-1}$: the stereographic variant slices via projection to $\mathbb{R}^{d-1}$; the linear variant slices via great-circle projections. Both yield $\mathcal{O}(n \log n)$ closed-form distances preserving geodesic geometry. (*ICLR 2025 Spotlight; ICML 2024 Spotlight*)
- Developed MinGSGW, a scalable GW approximation recasting the NP-hard quadratic assignment as a min-cost transport problem admitting efficient solvers, with theoretical bounds on the approximation gap. (*arXiv, 2026*)

Neural Simulators for Soft Tissue Deformation (*Ongoing*)

- Developed the Neural Augmented Kelvinlet (NAK): the closed-form Kelvinlet field $\mathbf{u}_\varepsilon$ serves as a physics prior; a network learns the residual $\hat{r}$, giving $\tilde{\mathbf{u}} = \mathbf{u}_\varepsilon + \hat{r}$. Trained on 20,800 FEM simulations with single- and multi-grasper support. (*AAAI 2026; U.S. Patent Application No. 63/998,405*)
- Leading development of SurgFormer, a transformer-based deformation model supporting resection-induced topological changes and real-time inference—capabilities absent in prior models limited to fixed-topology meshes. (*arXiv 2026*)

Selected Awards and Honors

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| ◦ Machine Learning Summer School (MLSS) NYC 2026, admitted with grant to cover program expenses, Columbia University | 2026 |
| ◦ Lambda Research Grant Program (\$2,000 in cloud compute credits) | 2025 |
| ◦ Google Cloud Research Credits (\$1,000 in cloud credits) | 2025 |
| ◦ Ranked Top 0.1% (335th of 140,000+) in Iran’s National University Entrance Exam | 2016 |

Academic and Professional Service

Reviewing: Provided review services for top-tier conferences and venues in ML and AI, including ICML (2025, 2026), ICLR (2025, 2026), NeurIPS (2025, 2026), AAAI (2026), ECCV (2026).

Mentoring: Zhanqi Zhu (Now a M.Sc student at USC),

WISE Summer Fellow, Vanderbilt University

Summer 2023

Professional Memberships:

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| ◦ Vanderbilt Lab for Immersive AI Translation (VALIANT), AI Scholar | 2025–Present |
| ◦ Association for Computing Machinery (ACM), Member | 2025–Present |
| ◦ Institute of Electrical and Electronics Engineers (IEEE), Member | 2025–Present |

Teaching Experience

Teaching Assistant — Geometric Deep Learning (CS, Graduate) Spring 2024
Vanderbilt University

Teaching Assistant — Statistics and Probability Spring 2021
Amirkabir University of Technology

Talks and Presentations

Geometric Neural Simulators for Modeling Liver Tissue Deformation (Oral) Dec 2024
13th Annual Symposium on Surgery, Intervention, and Engineering, Vanderbilt University

Additional Skills

Programming: Python, C, C++, Java

ML Libraries: PyTorch, Transformers, Diffusers, accelerate, JAX, scikit-learn, NumPy, SciPy

Systems & GPU: CUDA Runtime, CUTLASS, Nsight

Cloud & Tools: Linux, Git, Docker, AWS, GCP, Unity

Languages: English (Proficient), Persian (Native)